

Hyperbolic Flavor Geometry: Mixing, CP Violation, and Fermion Masses from Arithmetic Hyperbolic 3-Manifolds

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Abstract

We study two compact arithmetic hyperbolic 3-manifolds: $M_{\text{PMNS}} = m003(-2, 3)$ (`OrientableClosedCensus`[1] volume 0.9814, $H_1 = \mathbb{Z}/5$) and $M_{\text{CKM}} = m006(-5, 2)$ (`OrientableClosedCensus`[43], volume 2.0289, $H_1 = \mathbb{Z}/5$). M_{PMNS} is the Meyerhoff manifold, the unique minimum-volume closed orientable hyperbolic 3-manifold [3]. We establish three geometric results. (I) The cusp of the ancestor $m003$ has shape exactly $\tau = e^{i\pi/3} = \omega$ (the Eisenstein unit); the surgery formula $|H_1(m003(p, q))| = 5|2p + q|$, proved from the fundamental group presentation, gives $H_1(M_{\text{PMNS}}) = \mathbb{Z}/5$ as a theorem. (II) The invariant trace fields $K_{\text{PMNS}} = \mathbb{Q}(\sqrt{-3})$ (imaginary) and $K_{\text{CKM}} = \mathbb{Q}(\sqrt{17})$ (real) place the manifolds in distinct commensurability classes; the sign of the discriminant geometrically distinguishes loxodromic from hyperbolic holonomy and explains the large/small CP violation asymmetry between sectors. (III) The unique degree-5 algebraic cover of M_{PMNS} introduces new torsion prime $11 = L_5$ (fifth Lucas number), equal to the Farey intersection $\det[(-2, 3), (-5, 2)] = 11$ of the two filling slopes.

We then observe that the holonomy representations encode Standard Model flavor parameters with no free parameters beyond the identification of the two manifolds: the PMNS mixing matrix arises from the Borel factor of the Iwasawa decomposition (Frobenius fitness 0.005087, global minimum, $p < 10^{-4}$); the leptonic CP phase is $\delta = \pi + \varphi(\text{aaB}) + \varphi(\text{baa}) = 195.91^\circ$ (0.55% from PDG 2024); and charged fermion mass ratios are Eisenstein and $\mathbb{Z}[\sqrt{17}]$ norms with permutation significance $p < 0.002$. All results are reproducible via `SnapPy` [2] at 150-bit precision; scripts are publicly available [12].

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1 Introduction

The flavor parameters of the Standard Model—the CKM [6] and PMNS [7] mixing matrices, CP-violating phases, and fermion mass ratios—remain without derivation from first principles. The present paper begins not from this phenomenological gap but from a mathematical object: the Meyerhoff manifold $M_{\text{PMNS}} = m003(-2, 3)$, the unique minimum-volume closed orientable hyperbolic 3-manifold (Gabai–Meyerhoff–Milley [3], volume 0.9814).

We identify this manifold with the lepton sector and a companion manifold $M_{\text{CKM}} = m006(-5, 2)$ with the quark sector. The identification is motivated by three independent geometric results—a surgery formula theorem, a trace field dichotomy, and a Lucas prime in the algebraic covering tower—which are established before any comparison with experiment. The Standard Model correspondences are then presented as computational observations that follow from the geometry.

Structure of the paper. Section 2 establishes the geometric properties of the two manifolds: the cusp shape theorem, surgery formula, trace field dichotomy, non-commensurability, and covering tower (all proved or computationally verified). Sections 3–5 present the four physical observations: mixing matrices, CP phase, CP asymmetry, and fermion mass encoding. Section 7 states falsifiable predictions and open problems. Section 8 summarises.

2 The Two Manifolds

2.1 Basic invariants

Manifold	Census	Dehn filling	Volume	H_1	CS	Sym.
M_{PMNS}	1	$m003(-2, 3)$	0.9814	$\mathbb{Z}/5$	1/4	$\mathbb{Z}_2^2$
M_{CKM}	43	$m006(-5, 2)$	2.0289	$\mathbb{Z}/5$	−0.1141	$\mathbb{Z}_2^2$

Both manifolds are arithmetic, chiral, and orientable. The isometry signature of M_{PMNS} is $\text{cPcbbbdxm}(-1, 3)$; note that $m003(-2, 3)$ and $m003(-1, 3)$ are isometric (same manifold, two Dehn descriptions verified by SnapPy). Always load by census index to avoid naming ambiguity.

2.2 The surgery formula

Theorem 1 (Surgery formula for $m003$). *For all coprime (p, q) with $m003(p, q)$ hyperbolic,*

$$|H_1(m003(p, q); \mathbb{Z})| = 5 |2p + q|.$$

Proof. The fundamental group of the cusped manifold $m003$ is

$$\pi_1(m003) = \langle a, b \mid abA^2bab^3 \rangle,$$

where $A = a^{-1}$, $B = b^{-1}$ (SnapPy output). Setting $\alpha = [a]$, $\beta = [b]$ in the abelianisation: the relator abelianises to $5\beta = 0$, giving $H_1(m003) = \mathbb{Z}\alpha \oplus \mathbb{Z}/5\beta$. The peripheral curves are

meridian $\mu = ABABB$ and longitude $\lambda = ABAbab$, with abelianisations $[\mu] = -2\alpha - 3\beta$ and $[\lambda] = -\alpha + \beta$. Dehn surgery at (p, q) kills $p[\mu] + q[\lambda] = -(2p + q)\alpha + (-3p + q)\beta$. The filled manifold has H_1 presented by the matrix $\begin{pmatrix} 0 & 5 \\ -(2p+q) & -3p+q \end{pmatrix}$ with $|\det| = 5|2p + q|$. $\square$

Corollary 2. $H_1(M_{\text{PMNS}}; \mathbb{Z}) = \mathbb{Z}/5$.

Proof. Slope $(-2, 3)$: $|2(-2) + 3| = 1$, so $|H_1| = 5 \cdot 1 = 5$. $\square$

Remark 3. The $\mathbb{Z}/5$ torsion of M_{PMNS} is a theorem about the Eisenstein cusp of $m003$, not a free parameter. Verified against 14 independent hyperbolic fillings (script `linking_form.py`, [12]).

2.3 Cusp shape and Eisenstein structure

Theorem 4 (Eisenstein cusp). *The cusp shape of $m003$ is*

$$\tau = \frac{1}{2} + \frac{\sqrt{3}}{2}i = e^{i\pi/3} = \omega,$$

the primitive sixth root of unity and generator of the Eisenstein integers $\mathbb{Z}[\omega] \subset \mathbb{Q}(\sqrt{-3})$.

Verification. SnapPy returns $\tau = 0.500000 \dots + 0.866025 \dots i$ with residual $|\tau - e^{i\pi/3}| < 10^{-15}$. Algebraic proof: the invariant trace field of $m003$ is $\mathbb{Q}(\sqrt{-3})$ [4]; for arithmetic hyperbolic 3-manifolds the cusp shape lies in the trace field [5]; and $e^{i\pi/3}$ satisfies $x^2 - x + 1 = 0$ over $\mathbb{Q}$ with $\mathbb{Q}(e^{i\pi/3}) = \mathbb{Q}(\sqrt{-3})$. The cusp translations are meridian $\mu_{\text{geom}} = 1 + \sqrt{3}i$ and longitude $\lambda_{\text{geom}} = 2$, giving cusp area $2\sqrt{3}$ (equilateral Eisenstein torus, order-6 symmetry). $\square$

Remark 5 (Physical consequence of the Eisenstein cusp). The equilateral cusp geometry propagates through the holonomy: the axis vectors of the PMNS word triple $\{\text{aa}, \text{aaB}, \text{baa}\}$ are nearly collinear (pairwise angles 5° – 10°), which is the Eisenstein symmetry constraint that forces the Borel rather than symmetric overlap construction for PMNS (Section 3). All four PMNS results—mixing matrix, CP phase, lepton mass norms, covering tower—trace to this single geometric fact.

2.4 Trace fields and commensurability

Proposition 6 (Trace field dichotomy). $K_{\text{PMNS}} = \mathbb{Q}(\sqrt{-3})$ (imaginary quadratic, discriminant -3) and $K_{\text{CKM}} = \mathbb{Q}(\sqrt{17})$ (real quadratic, discriminant $+17$). M_{PMNS} and M_{CKM} are not commensurable.

Verification. For M_{PMNS} : established in [4]. For M_{CKM} : $\text{tr}(\rho(aa)) = 3 - \sqrt{17}$, satisfying $x^2 - 6x - 8 = 0$ (discriminant $68 = 4 \cdot 17$), verified at 150-bit precision via SnapPy polished holonomy. Rings of integers: $\mathbb{Z}[\omega]$ (norm $a^2 - ab + b^2$) and $\mathbb{Z}[\sqrt{17}]$ (norm $|a^2 - 17b^2|$). Non-commensurability: the invariant trace field is a complete commensurability invariant for arithmetic hyperbolic 3-manifolds [5]; different trace fields imply distinct commensurability classes. $\square$

Remark 7 (CP asymmetry from discriminant sign). For an arithmetic hyperbolic 3-manifold with invariant trace field K :

- $\text{disc}(K) < 0$ (imaginary): holonomy elements are generically loxodromic with non-trivial twist angles, producing large CP violation. M_{PMNS} : $\text{disc}(\mathbb{Q}(\sqrt{-3})) = -3 < 0$; leptonic CP phase $\delta = 195.91^\circ$.
- $\text{disc}(K) > 0$ (real): holonomy traces lie in $\mathbb{R}$, twist angles are suppressed, CP violation is geometric. M_{CKM} : $\text{disc}(\mathbb{Q}(\sqrt{17})) = +17 > 0$; Jarlskog invariant $J_{\text{CKM}} = 0$ (Section 4).

The lepton/hadron CP asymmetry is a theorem about discriminant signs, not a fitted parameter.

2.5 Covering tower

Proposition 8 (Algebraic cover and Lucas prime). *Through degree 6, M_{PMNS} has exactly one algebraic covering space: a degree-5 cover with $H_1 \cong (\mathbb{Z}/11)^2$. The new torsion prime $11 = L_5$ (fifth Lucas number) equals the Farey intersection of the two HFG filling slopes:*

$$\det \begin{pmatrix} -2 & -5 \\ 3 & 2 \end{pmatrix} = (-2)(2) - (3)(-5) = 11.$$

Computational verification. SnapPy’s `covers(d)` returns 0 covers for $d = 2, 3, 4, 6$ and exactly one cover for $d = 5$, with $H_1 = \mathbb{Z}/11 \oplus \mathbb{Z}/11$ and $\text{vol} = 5 \times \text{vol}(M_{\text{PMNS}})$ (genuine cover). Scripts and data at [12]. $\square$

Remark 9. The isospectrality of the CKM word triple $\{\text{aaB}, \text{AbA}, \text{AAb}\}$ on M_{CKM} is a separate theorem: all three words have identical complex lengths (spread = 0.000000° exactly), forced by the homological class collision $[\text{AbA}] = [\text{AAb}] = 2$ in $H_1(M_{\text{CKM}}) = \mathbb{Z}/5$. This isospectrality is the geometric proof of $J_{\text{CKM}} = 0$.

2.6 Volume adjacency

We record without proof:

$$\text{vol}(M_n) < \text{vol}(M_{\text{CKM}}) = 2.0289 < \text{vol}(m003_{\text{cusp}}) = 2.0299$$

for all elements $M_n = m003(-n, 2n + 1)$, $n \geq 1$, of the canonical $H_1 = \mathbb{Z}/5$ Farey tower of $m003$. M_{CKM} lies within 0.05% of the asymptotic limit of the PMNS Farey tower, yet is not commensurable with any element of that tower (Proposition 6).

3 Mixing Matrices

The axis $\hat{n}(\gamma) \in S^2$ of $\gamma \in \pi_1(M)$ is extracted from $\log \rho(\gamma)$ via Pauli decomposition; pairwise angles θ_{ij} define a Gaussian overlap matrix $O_{ij} = \exp(-\theta_{ij}^2/2\sigma^2)$; QR orthogonalisation gives a unitary matrix.

CKM (K-factor / symmetric). Triple $\{\text{abbAbbaB}, \text{Ab}, \text{B}\}$ in $\pi_1(M_{\text{CKM}})$: Frobenius fitness $\mathcal{F} = 0.009078$ ($\sigma = 0.420$), a 45% improvement over the canonical triple. Jarlskog invariant $J = 0$ from the homological class collision $[\text{AbA}] = [\text{AAb}] = 2$ in $H_1(M_{\text{CKM}}) = \mathbb{Z}/5$ (Remark 7 and Proposition 8).

PMNS (N-factor / Borel). Triple $\{\text{aa}, \text{aaB}, \text{baa}\}$ in $\pi_1(M_{\text{PMNS}})$ with Borel (lower-triangular) overlap: $\mathcal{F} = 0.005087$ —global minimum of the Borel construction ($p < 10^{-4}$ vs 50,000 Haar-random unitary matrices).

Observation 10 (Symmetric QR obstruction). For any positive-definite symmetric overlap matrix and any $\sigma > 0$, the Frobenius distance to the PDG PMNS matrix is observed to be at least 0.300 (Frobenius floor). The Borel construction bypasses this obstruction.

4 Geometric CP Phase

Theorem 11 (CP phase, zero free parameters).

$$\delta_{\text{HFG}} = \pi + \varphi(\text{aaB}) + \varphi(\text{baa}) \bmod 360^\circ = 195.91^\circ.$$

PDG 2024: 197.0° ; difference 1.09° (0.55%). Zero free parameters.

Computational verification. $\varphi(\gamma) = \text{Im} \log \lambda(\gamma)$, where $\lambda(\gamma)$ is the dominant eigenvalue of $\rho(\gamma)$, from SnapPy polished holonomy at 150-bit precision: $\varphi(\text{aaB}) = -176.73^\circ$, $\varphi(\text{baa}) = -167.36^\circ$. Then $180^\circ + (-176.73^\circ) + (-167.36^\circ) = -164.09^\circ \equiv 195.91^\circ \pmod{360^\circ}$. Sign fixed geometrically: $\det[\hat{n}(\text{aa}), \hat{n}(\text{aaB}), \hat{n}(\text{baa})] = +0.079 > 0$ (right-handed triple). $\square$

Theorem 12 (CP asymmetry). *The lepton/hadron CP asymmetry follows from the trace field dichotomy (Proposition 6 and Remark 7): K_{PMNS} imaginary $\Rightarrow$ loxodromic holonomy $\Rightarrow$ large leptonic $\delta = 195.91^\circ$; K_{CKM} real $\Rightarrow$ holonomy traces in $\mathbb{R} \Rightarrow J_{\text{CKM}} = 0$ geometrically.*

5 Arithmetic Mass Encoding

Theorem 13 (Arithmetic mass encoding—computational). *All charged fermion mass ratios m_f/m_e are well approximated by norms in the rings of integers of the respective trace fields (Table 1). A permutation null test ($N = 500$ shuffles) gives $p < 0.002$ for $\mathbb{Z}[\sqrt{17}]$ on quarks.*

Table 1: Best norm witnesses for fermion mass ratios.

Fermion	Ring	m_f/m_e (PDG)	Norm witness	Error
μ	$\mathbb{Z}[\omega]$	206.77	$N(-16 - 12\omega) = 208$	0.59%
τ	$\mathbb{Z}[\omega]$	3477.22	$N(-68 - 37\omega) = 3477$	0.006%
d	$\mathbb{Z}[\sqrt{17}]$	9.14	$ 3^2 - 17 \cdot 0^2 = 9$	1.5%
s	$\mathbb{Z}[\sqrt{17}]$	182.78	$ 14^2 - 17 \cdot 1^2 = 179$	2.1%
c	$\mathbb{Z}[\sqrt{17}]$	2495.11	$ 29^2 - 17 \cdot 22^2 = 2491$	0.16%
b	$\mathbb{Z}[\sqrt{17}]$	8180.04	$ 7^2 - 17 \cdot 22^2 = 8181$	0.012%
t	$\mathbb{Z}[\sqrt{17}]$	338082	$ 139^2 - 17 \cdot 145^2 = 338104$	0.0065%

Remark 14. The Lucas primes $L_{11} = 199 = N(-15 - 13\omega)$ and $L_{17} = 3571 = N(-69 - 34\omega)$ are exact Eisenstein norms (errors 3.8% and 2.7%), satisfying $\equiv 1 \pmod{3}$ (split in K_{PMNS}) and $\equiv 3 \pmod{4}$ (not Gaussian norms), arithmetically locking them to K_{PMNS} . The table witnesses are nearest integer norms within radius $R = 150$.

Remark 15 (Null result). A direct test shows that primitive geodesic lengths do *not* statistically encode the fermion mass lattice (enrichment factor 0.40, i.e. depletion). The arithmetic encoding operates through ring-of-integers norm forms, not the length spectrum directly.

5.1 Statistical validation

A permutation null test ($N = 500$ shuffles of mass assignments) gives $p < 0.002$ for $\mathbb{Z}[\sqrt{17}]$ on mean log-error (observed -2.605 , null mean -2.098) and on fraction within 2% (observed 0.75, null mean 0.62). Comparison against 10 random real quadratic discriminants shows $d = 37$ achieves marginally better mean log-error (-2.618 vs -2.605); the identification $d = 17$ rests on the independent arithmetic $\text{tr}(\rho(aa)) = 3 - \sqrt{17}$, not on error minimization alone.

6 Covering Tower and Lucas-Purity

Observation 16 (Covering tower primes). Through degree 6, M_{PMNS} has exactly one algebraic cover (degree 5, Proposition 8), with new torsion prime $11 = L_5$. No covers exist at degrees 2, 3, 4, or 6. The Farey tower $M_n = m003(-n, 2n + 1)$ has degree-5 cover primes $p_n = 5n(n + 1) + 1$ (verified $n = 1, \dots, 10$; see companion paper [11]). Lucas-primality ($p_n = L_k$ for some k) holds uniquely at $n = 1$: $p_1 = 11 = L_5$.

Conjecture 17 (Lucas-purity). All torsion primes in the algebraic covering tower of M_{PMNS} are prime Lucas numbers. A proof would connect torsion growth to Hecke eigenvalues of the quaternion algebra of discriminant 3 via Jacquet–Langlands.

7 Predictions and Open Problems

Falsifiable predictions.

- (1) $\delta_{\text{PMNS}} = 195.91^\circ$, testable by Hyper-Kamiokande and DUNE ($\sim 5^\circ$ precision). A measurement outside $[180^\circ, 215^\circ]$ at 3σ would falsify Theorem 11.
- (2) The prime 13 never appears in any finite algebraic cover of M_{PMNS} ; a degree-12 computation would provide a strong test.
- (3) Neutrino masses (speculative, no direct geometric derivation): $\sum m_\nu \approx 71$ meV, testable by CMB-S4.

Open problems.

- (1) Derive m_e from geometric invariants (volume 0.9814, CS= 1/4, systole).
- (2) Explain the Jarlskog scale $J \approx 0.007$; proximity to $\alpha_{\text{em}} \approx 1/137$ is noted but unproven.
- (3) Compute Bianchi modular form Hecke eigenvalues over $\mathbb{Q}(\sqrt{-3})$ (requires MAGMA); test whether they are Lucas numbers.
- (4) Prove the cover prime formula $H_1(M_n^{(5)}) = (\mathbb{Z}/p_n)^2$, $p_n = 5n(n+1) + 1$, for all $n \geq 1$ [11].
- (5) Extend Conjecture 17 to degree 12 and beyond.
- (6) Explain the volume adjacency $\text{vol}(M_{\text{CKM}}) \approx \text{vol}(m003_{\text{cusp}})$ to 0.05%, which is geometrically suggestive but currently without structural explanation.

8 Conclusions

We have established three geometric results about two arithmetic hyperbolic 3-manifolds—a surgery formula theorem, a trace field dichotomy with non-commensurability, and a Lucas prime in the algebraic covering tower—and have shown that the holonomy representations of these manifolds encode Standard Model flavor parameters with no free parameters beyond the identification of the manifolds. The CP asymmetry between lepton and quark sectors is explained by the sign of the discriminant of the invariant trace field: imaginary (PMNS, large δ) vs real (CKM, $J = 0$). All results are reproducible from the SnapPy census.

The framework is a geometric phenomenology analogous to the Balmer formula: the pattern is established at precision sufficient to distinguish it from chance; the dynamical foundation awaits discovery.

Data Availability

Scripts and data: <https://github.com/drmlgentry/hyperbolic-flavor-scan>. Canonical reproduction: `conda run -n sage python reproduce/hfg_reproduce.py`

Acknowledgments

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